

UNITED STATES OF AMERICA
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the
Interstate Transmission System
Docket No. RM26-4-000

**COMMENTS OF INDEPENDENT INFRASTRUCTURE RESEARCHERS ON HIGH-
RESOLUTION TELEMETRY AND OSCILLATION MONITORING FOR LARGE
COMPUTATIONAL LOADS UNDER FPA § 215**

Pursuant to the Federal Energy Regulatory Commission's ("Commission" or "FERC") continuing invitation to file comments in the above-captioned docket, and in light of the Commission's June 18, 2026 orders directing each Regional Transmission Organization and Independent System Operator to justify or reform its large-load interconnection practices under Section 206 of the Federal Power Act, The Computational Infrastructure & Grid Telemetry Initiative ("Commenters") respectfully submit these comments.

I. STATEMENT OF INTEREST

Commenters are independent computational infrastructure and energy systems researchers. Commenters analyze publicly available regional transmission interconnection data, NERC reliability proceedings, and the electrical transient characteristics of high-density computational loads. Commenters submit these comments to assist the Commission in developing an administrable, reliability-based framework under Section 215 of the Federal Power Act in Docket No. RM26-4-000.

Commenters have analyzed publicly available interconnection queue data, NERC reliability filings, and related technical literature concerning the integration of large computational loads — including AI training clusters — into the bulk power system. Commenters submit these comments to assist the Commission in developing a technically grounded, administrable framework for monitoring such loads under its existing reliability authority.

II. SUMMARY OF POSITION

Commenters urge the Commission to: (1) direct the Electric Reliability Organization to incorporate specific high-resolution telemetry and oscillation-detection requirements for large computational loads into the Reliability Standard under development in NERC Project 2026-02; (2) support development of a Critical Energy/Electric Infrastructure Information ("CEII")-protected protocol enabling the Electric Reliability Organization to share defined categories of load-verification data with the Department of Commerce for the limited purpose of semiconductor export-diversion triage; and (3) confirm that this proposal is grounded exclusively in the Commission's Federal Power Act Section 215 reliability authority and does not extend to, or

depend upon, resolution of the separate and pending questions concerning the Commission's jurisdiction over large-load interconnection and retail ratemaking.

III. ARGUMENT

A. The Requested Telemetry and Monitoring Requirements Are Grounded in the Commission's Well-Established Section 215 Reliability Authority

Section 215 of the Federal Power Act, 16 U.S.C. § 824o, authorizes the Commission to certify an Electric Reliability Organization and to approve mandatory, enforceable Reliability Standards governing the bulk power system. This authority is long-settled and does not depend on, or implicate, the separate question — currently the subject of active dispute among stakeholders in this docket — of whether and to what extent the Commission may assert jurisdiction over the interconnection of retail loads under Sections 205 and 206 of the Federal Power Act. Commenters take no position in these comments on that separate jurisdictional question.

The requests herein rest solely on the Commission's undisputed authority to direct the Electric Reliability Organization to develop Reliability Standards addressing bulk power system reliability risks — an authority the Commission has already exercised with respect to computational loads specifically. On July 16, 2026, acting on its own motion under Section 215(d)(5) of the Federal Power Act, the Commission directed NERC to develop Reliability Standards, and associated Glossary definitions, addressing computational load integration, together with Rules of Procedure revisions establishing registry criteria for a new “computational load entity” category, both due by December 31, 2026, and a Phase II workplan due March 1, 2027. Order Directing NERC to Develop Reliability Standards Pertaining to Computational Load Integration, Docket No. RD26-7-000, 196 FERC ¶ 61,031 (July 16, 2026). That process, NERC Project 2026-02, has since produced a draft Reliability Standard, CLO-001-1 (Computational Load Interconnection, Studies, & Modeling Data), together with related revisions to FAC-001-5 and FAC-002-5. As of this filing, these remain in draft form, with additional balloting anticipated in fall 2026 and possible NERC Board of Trustees adoption by the end of 2026.

B. NERC Has Already Documented the Reliability Basis for High-Resolution Telemetry and Oscillation Monitoring

NERC's Reliability and Security Technical Committee established a Large Load Task Force to analyze the reliability impacts of emerging large loads, including data centers used for AI training. NERC's resulting technical work describes AI training clusters as exhibiting rapid load fluctuations during training operations and, in particular, during periodic checkpoint operations — a characteristic distinct from the load behavior of other large-load categories. Separately, NERC issued a Level 3 Essential Action alert following documented bulk power system events in which more than 1,000 megawatts of computational load disconnected within seconds, and NERC has identified sub-synchronous oscillation events associated with computational facilities in more than one Interconnection.

These are precisely the categories of documented, recurring reliability events that Section 215 Reliability Standards exist to address.

C. A Two-Part Load Signature, Applied as a Rebuttable Presumption, Identifies Covered Facilities Without Requiring the Commission or NERC to Classify the Nature of a Facility's Computing Activity

Commenters recognize that the Commission and NERC are not, and should not be, in the business of classifying what a given facility is computing. Commenters therefore propose a two-part electrical signature, keyed entirely to observable grid behavior, to identify facilities warranting the proposed telemetry requirements:

- (1) a sustained capacity factor materially higher, and diurnal and seasonal variance materially lower, than comparable industrial loads, measured over a rolling multi-week period; and
- (2) synchronized, large-magnitude power transients correlated with periodic intervals consistent with distributed-training checkpoint operations, superimposed on the baseline described in (1).

Commenters propose that this signature operate as a rebuttable presumption: a facility meeting both criteria would be presumptively subject to the proposed telemetry requirement, but could rebut that presumption with load-factor and process documentation demonstrating that its demand pattern reflects a different industrial process, such as continuous-process manufacturing or hydrogen electrolysis. This signature is intended to identify large, synchronized training operations specifically, consistent with the load behavior NERC has already documented, and is not intended to, and would not, reach inference-serving or other more variable, demand-responsive computing infrastructure.

Commenters do not propose specific numerical thresholds for capacity factor or diurnal variance in these comments. Consistent with NERC's standard-setting process, Commenters recommend that the Standards Drafting Team for CLO-001-1 establish these thresholds using the operating data NERC and the Regional Transmission Organizations will collect through the registration and reporting process directed in Docket No. RD26-7-000, rather than have the Commission fix specific numerical values by rule in the first instance.

D. The Commission Should Support Development of a CEII-Protected Protocol for Sharing Defined Load-Verification Data With the Department of Commerce for Export-Diversion Triage

Data collected under a Section 215 Reliability Standard is properly protected as Critical Energy/Electric Infrastructure Information under 18 C.F.R. § 388.113 and is not, without more, available for use by other agencies for other purposes. Commenters do not ask the Commission to alter that protection. Rather, Commenters ask the Commission to support development of a defined, CEII-compliant protocol under which NERC may share specified load-verification data with the Department of Commerce's Bureau of Industry and Security ("BIS"), on a need-to-know basis, solely for use as one triage input supporting BIS's existing semiconductor end-use and diversion-verification responsibilities.

This proposal is directly responsive to the interagency coordination priorities identified in the Administration's July 2025 AI Action Plan, which calls for a Department of Commerce-led enforcement collaboration to monitor emerging AI compute risks and expand end-use monitoring. Commenters emphasize that this is a request to establish an information-sharing channel supporting an existing enforcement mission, not a request that the Commission or NERC create

any new mandatory disclosure regime directed at domestic AI development. A corresponding petition asking BIS to establish its side of this protocol is being filed concurrently with the Bureau.

E. This Proposal Concerns Bulk Power System Reliability Telemetry Only and Does Not Present the Retail Ratemaking and State-Jurisdiction Concerns Raised Elsewhere in This Docket

Commenters are aware that other parties to this docket, including the National Association of Regulatory Utility Commissioners, have raised significant concerns regarding the extent of the Commission's jurisdiction over retail load interconnection and the potential for federal action in this docket to intrude on matters of state retail ratemaking authority. Commenters' proposal is expressly limited to bulk power system reliability telemetry under Section 215 and does not propose any change to retail rate design, cost allocation, or the retail interconnection process, and takes no position on the separate Section 205/206 jurisdictional questions pending elsewhere in this docket.

Commenters submit that a reliability-standard-based approach of the kind proposed here is materially less likely to raise the jurisdictional concerns identified by state regulators than an approach grounded in the Commission's ratemaking authority. Commenters recognize, however, that this question has not been definitively resolved and do not ask the Commission to so hold in this proceeding.

IV. REQUESTED ACTION

For the foregoing reasons, Commenters respectfully request that the Commission:

1. Direct NERC to consider, in NERC Project 2026-02, Reliability Standard requirements incorporating high-resolution telemetry and oscillation-detection reporting for large computational loads meeting the criteria described in Section III.C above;
2. Support NERC's development, in coordination with the Department of Commerce, of a CEIL-compliant protocol for sharing defined load-verification data for the limited purpose described in Section III.D above; and
3. Confirm that any resulting Reliability Standard will be limited to bulk power system reliability purposes and will not resolve, or be construed to resolve, the separate jurisdictional questions concerning retail load interconnection pending elsewhere in this docket.

V. CONCLUSION

For the reasons stated above, Commenters respectfully request that the Commission act consistent with these comments in Docket No. RM26-4-000 and in NERC Project 2026-02.

Respectfully submitted,

/s/ Technical Drafting Committee

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